

I’m Fine, But My Voice Isn’t: Cross-Modal Affective Dissonance Detection for Reflective Journaling

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Abstract

Digital journaling creates an authenticity gap: users consciously translate raw emotions into text, often sanitizing narratives even in private writing. We formalize this as **Cross-Modal Affective Dissonance Detection** (CADD), a directional three-way classification distinguishing Masking (positive text, negative acoustics), Coping (negative text, positive acoustics), and Congruent utterances, grounded in Gross’s process model of emotion regulation. We present three further contributions: (i) **CADD-Journal**, a 1,800-sample TTS dataset with a shared-sentence-pool design that provably isolates acoustic signal from textual content; (ii) **DACM**, a dual-encoder model with asymmetric cross-modal attention that resolves a gradient degeneracy in pooled fusion, achieving macro-F1 0.711 ± 0.025 —with a four-step ablation demonstrating that asymmetric attention is the dominant driver (+0.242) while the DIM is effective only on cross-modal features (+0.033); and (iii) a **domain gap quantification**: zero-shot evaluation across three naturalistic corpora reveals a substantial gap between TTS-trained models and real speech, and we identify two concrete requirements for future in-the-wild corpus construction. **ReflectJournal**, a proof-of-concept iOS application, operationalizes the framework and provides a deployment platform for naturalistic data collection.¹

CCS Concepts

• **Computing methodologies** → **Neural networks**; *Natural language processing*; • **Human-centered computing** → *Human computer interaction (HCI)*.

Keywords

affective computing, emotion regulation, multimodal interaction, speech–text dissonance, digital journaling, cross-modal attention

1 Introduction

Expressive writing produces measurable benefits for physical and mental health when individuals engage authentically with their emotional experience [12, 13]. Yet impression management persists even in private journaling [10]: users subconsciously sanitise narratives to maintain a positive self-image [9], and self-concealment predicts anxiety and depression. Current journaling applications rely exclusively on text, providing no mechanism to detect or surface this authenticity gap.

Voice as a hidden signal. Prosodic features—pitch, energy, speaking rate, and timbre—reliably convey affective states that verbal content alone cannot capture [5, 7]. Crucially, vocal and

verbal channels frequently diverge; in IEMOCAP [1] and CMU-MOSEI [19], acoustic emotion regularly contradicts lexical content.

The lexical dominance problem. Existing multimodal models assume cross-modal alignment and consequently suppress the dissonance signal. Recent work shows that audio–language models rely on lexical semantics under acoustic–semantic conflict while underutilising prosodic cues—a phenomenon termed *lexical dominance* [2]. Prior fusion methods progress from tensor products [18] to directional cross-modal transformers [14] and contrastive alignment [11, 15], but typically treat cross-modal divergence as noise to minimise rather than a signal to exploit. CADD reframes this failure mode as a detection target.

Specifically, our contributions are:

- (1) **CADD task**: three-way classification distinguishing Masking (text+, audio−), Coping (text−, audio+), and Congruent (aligned), grounded in Gross’s process model [8].
- (2) **CADD-Journal**: a 1,800-sample TTS dataset whose shared-sentence-pool design ensures any F1 advantage is genuinely multimodal.
- (3) **DACM architecture**: a dual-encoder model with asymmetric cross-modal attention resolving a gradient degeneracy in pooled fusion, and a Dissonance Interaction Module (DIM) producing an interpretable mismatch score S .
- (4) **Domain gap quantification**: zero-shot evaluation across three naturalistic corpora reveals a substantial gap between TTS dissonance and real speech, providing, to our knowledge, the first systematic quantification of this gap, together with two concrete requirements for future naturalistic corpus construction. **ReflectJournal**, an iOS design probe, operationalizes the framework and provides a deployment platform for that collection.

2 Task Definition

Given (T_i, A_i) —a text transcript and speech recording—CADD requires $f_\theta : (T, A) \rightarrow \{0, 1, 2\}$ over $\mathcal{D} = \{(T_i, A_i, y_i)\}_{i=1}^N$. Table 1 defines the taxonomy drawing on Gross’s process model [8] as a conceptual lens; labels are interpretive anchors rather than claims that TTS-rendered samples instantiate genuine emotion regulation. The directional distinction is therapeutically meaningful: Masking and Coping warrant different reflective prompts. A text-only classifier collapses to predicting Congruent under shared-sentence-pool conditions, validating that any F1 gain must exploit acoustic information. Consequently, we use macro-F1 as our primary evaluation metric.

¹Code: <https://github.com/cirresearch/CADD>

Table 1: CADD affective taxonomy.

Class	Condition	Regulation	Label
Masking	text+, audio−	Suppression	0
Coping	text−, audio+	Reappraisal	1
Congruent	aligned	—	2

3 DACM

3.1 Architecture Overview

We name our model DACM (Dissonance-Aware Cross-Modal model). Figure 1 illustrates four sequential stages: (1) unimodal encoding with Weighted Layer Pooling (WLP), (2) pre-attention auxiliary heads, (3) Asymmetric Cross-Modal Attention, and (4) the Dissonance Interaction Module (DIM).

3.2 Unimodal Encoders

Text is encoded by XLM-RoBERTa-base [4], producing pooled CLS vector $\mathbf{h}_t \in \mathbb{R}^{768}$ and full token sequence $\mathbf{H}_t$. Audio is encoded by WavLM-base-plus [3], producing masked-mean vector $\mathbf{h}_a \in \mathbb{R}^{768}$ and frame sequence $\mathbf{H}_a$. WLP [16] computes a learnable weighted average across all 12 hidden layers; both encoders are frozen, ensuring identical encoder capacity across all ablation conditions.

3.3 Asymmetric Cross-Modal Attention

Standard pooled-to-pooled attention is degenerate. Specifically, the attention score reduces to a scalar, $\text{softmax}(\cdot) \equiv 1$, so W_Q and W_K receive zero gradient. We resolve this by using one modality’s pooled vector as query and the opposing modality’s full sequence as keys and values:

$$\begin{aligned}\mathbf{h}_{t \rightarrow a} &= \text{MHA}(Q=\mathbf{h}_t, K=V=\mathbf{H}_a), \\ \mathbf{h}_{a \rightarrow t} &= \text{MHA}(Q=\mathbf{h}_a, K=V=\mathbf{H}_t).\end{aligned}$$

The score tensor $[B \times H \times 1 \times T']$ yields a non-trivial softmax and meaningful gradients for both W_Q and W_K . Unlike MulT [14], which uses full sequence-to-sequence cross-attention, our pooled-query design is the specific choice that eliminates the scalar-softmax degeneracy.

3.4 Dissonance Interaction Module

Let $\mathbf{h}'_t = \mathbf{h}_{t \rightarrow a}$ and $\mathbf{h}'_a = \mathbf{h}_{a \rightarrow t}$, both in $\mathbb{R}^{768}$. The DIM computes four features:

$$\begin{aligned}\mathbf{g} &= \sigma(W_g[\mathbf{h}'_t; \mathbf{h}'_a]) \odot (\mathbf{h}'_t - \mathbf{h}'_a), \quad \mathbf{m} = \mathbf{h}'_t \odot \mathbf{h}'_a, \\ s &= \cos(\mathbf{h}'_t, \mathbf{h}'_a), \quad \mathbf{d} = \text{sg}(\hat{y}_t) - \text{sg}(\hat{y}_a),\end{aligned}$$

with $\mathbf{g}, \mathbf{m} \in \mathbb{R}^{768}$, $s \in \mathbb{R}$, and $\mathbf{d} \in \mathbb{R}^3$ (unimodal class-logit difference); $\text{sg}(\cdot)$ denotes stop-gradient, ensuring $\mathbf{d}$ captures genuine unimodal disagreement rather than a back-propagated artefact of $\mathcal{L}_{\text{CE}}$. The **mismatch score** $S = \frac{1-s}{2} \in [0, 1]$ ($S \rightarrow 1$ maximally dissonant). The concatenated feature $\mathbf{f} = [\mathbf{g}; \mathbf{m}; s; \mathbf{d}] \in \mathbb{R}^{1540}$ (i.e. $768 + 768 + 1 + 3$) feeds a two-layer MLP (hidden 512, dropout 0.4) and a binary agreement head.

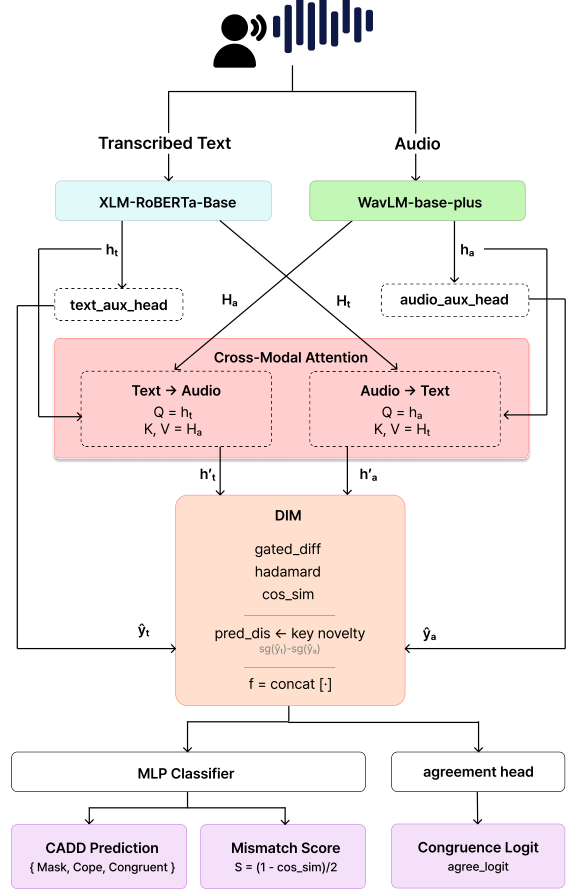


Figure 1: DACM architecture. Frozen text (blue, XLM-RoBERTa) and audio (green, WavLM) encoders feed Asymmetric Cross-Modal Attention (pink), in which each modality’s pooled vector queries the opposing modality’s full sequence; this design resolves the scalar-softmax gradient degeneracy of pooled-to-pooled attention (§3.3). The Dissonance Interaction Module (orange) computes gated difference, Hadamard product, cosine similarity, and a stop-gradient unimodal-disagreement vector, producing the CADD class label and the interpretable mismatch score $S \in [0, 1]$ (purple).

3.5 Training Objective

$$\begin{aligned}\mathcal{L} &= \mathcal{L}_{\text{CE}}(\varepsilon=0.1) + 0.3 \mathcal{L}_{\text{margin}} \\ &\quad + 0.2 \mathcal{L}_t^{\text{aux}} + 0.2 \mathcal{L}_a^{\text{aux}} + 0.1 \mathcal{L}_{\text{agr}},\end{aligned}\quad (1)$$

where $\mathcal{L}_{\text{margin}}$ is a bipolar cosine margin loss on L2-normalised unimodal projections (pulling congruent pairs together, pushing dissonant pairs apart); scalar weights were selected by coarse grid search.

4 CADD-Journal Dataset

Each audio file is rendered by ElevenLabs Eleven v3 [6] with an explicit emotion tag (negative: *sad, angry, annoyed, appalled*; positive: *happy, excited, cheerful, calm*), providing acoustic control unavailable in naturalistic speech. The shared-sentence-pool design renders 100 seed sentences across all acoustic conditions, so text content is uninformative for label prediction—any F1 advantage is genuinely multimodal. Splits are stratified at the sentence_id level; for each dissonance class, 2 emotion tags are randomly drawn per sentence–voice pair from a pool of 4, yielding $100 \times 3 \times 2 = 600$ samples per class across 3 voices (Jarnathan, Juniper, Eve); see Table 2.

Table 2: CADD-Journal statistics.

Class	Train	Val	Test	Total
Masking	420	90	90	600
Coping	420	90	90	600
Congruent	420	90	90	600
Total	1260	270	270	1800

5 Experiments

5.1 Setup

DACM-Base replaces asymmetric attention with simple pooled fusion ($[\mathbf{h}_t - \mathbf{h}_a; \mathbf{h}_t \odot \mathbf{h}_a]$), holding all other components constant to isolate the attention contribution. **DACM-noAttn** retains auxiliary heads and the DIM but removes asymmetric attention, reverting to pooled fusion ($\mathbf{h}_t, \mathbf{h}_a$) as the DIM’s input; this isolates the attention contribution while holding auxiliary heads and DIM presence constant. **DACM-noDIM** applies asymmetric cross-modal attention but replaces the DIM with direct concatenation of $\mathbf{h}'_t$ and $\mathbf{h}'_a$, isolating the DIM’s contribution on cross-modal features. All models use frozen encoders, AdamW ($\beta=0.9, 0.999$), lr 5×10^{-4} , batch 16, patience 7, gradient clipping 1.0, and a static 70/15/15 split. All results are mean $\pm$ std over seeds {42, 123, 456}.

5.2 Results on CADD-Journal

Table 3: Ablation on CADD-Journal test set (3 seeds each). Per-class F1 at best seed; Macro = mean $\pm$ std. [†]TextOnly degenerates to predicting only Congruent under shared-sentence-pool conditions; stratified random guessing yields macro-F1 = 0.333. [‡]noAttn: asymmetric attention removed; DIM operates on raw pooled $\mathbf{h}_t, \mathbf{h}_a$.

Model	Mask.	Cop.	Cong.	Macro
TextOnly [†]	0.000	0.000	0.500	0.167 ± 0.000
AudioOnly	0.630	0.535	0.255	0.448 ± 0.023
DACM-Base	0.628	0.562	0.183	0.441 ± 0.014
DACM-noAttn [‡]	0.642	0.578	0.254	0.469 ± 0.021
DACM-noDIM	0.794	0.719	0.548	0.678 ± 0.007
DACM (ours)	0.839	0.845	0.522	0.711± 0.025

TextOnly degenerates to predicting a single class (Congruent), yielding macro-F1 = 0.167. Note that stratified random guessing achieves macro-F1 = 0.333 under our balanced 3-class split; collapsing to one class is strictly worse because two of the three class-wise F1 scores are forced to zero. This confirms that text content alone carries no class-discriminative signal under the shared-sentence-pool design, so any improvement above this floor must exploit acoustic information. DACM-Base (0.441) falls below AudioOnly (0.448), confirming that simple pooled fusion actively suppresses the audio signal—consistent with lexical dominance under shared-sentence-pool conditions [2].

The four-step attribution ladder isolates each component’s contribution. Adding auxiliary heads and the DIM on top of pooled fusion (DACM-noAttn, 0.469) yields only $\Delta=+0.028$ over DACM-Base: the DIM is nearly inert when operating on uncontextualised unimodal features. Replacing pooled fusion with asymmetric cross-modal attention — which provides semantically contextualised features for the DIM to operate on — accounts for the dominant gain ($\Delta=+0.242, 0.469 \rightarrow 0.711$). The DIM then adds a further $+0.033$ on top of attention-contextualised features (DACM vs. DACM-noDIM), consistent with it being architecturally motivated for inference (mismatch score S) rather than as the primary classification driver. Gains concentrate on Masking and Coping—the two classes requiring genuine cross-modal reasoning. A leave-one-voice-out (LOVO) evaluation yields mean macro-F1 0.655 (Jarnathan: 0.645, Juniper: 0.689, Eve: 0.632), only 0.056 below in-domain, confirming voice-independent dissonance signatures.

5.3 Domain Gap: From Controlled TTS to Real Speech

Table 4: Zero-shot cross-corpus evaluation of best DACM checkpoint (in-domain F1 0.736). [†]Silver labels via SiBERT + emotion-to-valence mapping. [‡]2-class (Masking/Coping); random 0.500. [§]3-class; random 0.333. CH-SIMS uses dataset gold CADD-mapped labels.

Dataset	Lang.	Size	Label	F1
CADD-Journal	EN	1800	gold	0.736
CMU-MOSEI	EN	572	silver [†]	0.512 [‡]
IEMOCAP	EN	5674	silver [†]	0.362 [§]
CH-SIMS [17]	ZH	4404	gold	0.335 [§]
Random (2-cl / 3-cl)				0.500 / 0.333

Table 4 is the paper’s most practically significant finding: a model that achieves 0.711 macro-F1 under controlled conditions collapses to near random on all three naturalistic corpora (CMU-MOSEI 0.512 over a 0.500 baseline; IEMOCAP 0.362; CH-SIMS 0.335). This reveals a *severe and previously unquantified* domain gap between TTS-synthesised dissonance and naturalistic human speech. The gap is not merely a model limitation—it reflects the absence of affective corpora targeting self-reflective monologues with concealed emotion. Closing it requires corpora built on two concrete constraints: (1) independent dual-channel annotation separating text sentiment

from acoustic valence, and (2) prospective user-consented labelling, as concealed states cannot be annotated reliably post-hoc.

6 ReflectJournal

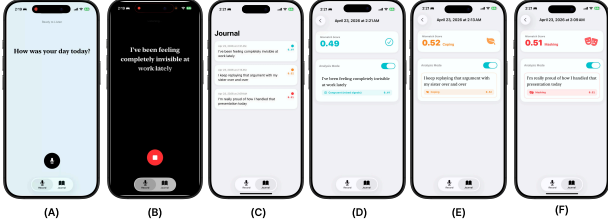


Figure 2: ReflectJournal: (A/B) recording interface, (C) journal list with mismatch scores, (D/E/F) detail view for congruent, coping, and masking states respectively. “Congruent” denotes text–audio alignment per Table 1.

ReflectJournal is an iOS design probe that operationalizes the CADD framework (Figure 2). The recording interface (A/B) withholds metrics during capture to minimise the observer effect; detailed analysis is available on-demand via a toggle in the journal list (C). The detail view (D/E/F) surfaces the mismatch score S and class-specific reflective prompts when $S > 0.05$, inviting self-inquiry based on detected cross-modal divergence. Beyond its journaling function, ReflectJournal serves as a deployment platform for future in-the-wild corpus collection with user-consented affective labels, providing a direct pathway to bridge the TTS-to-real-speech gap identified in §5.3.

7 Discussion and Conclusion

The central finding of this work is the domain gap: a model achieving 0.711 macro-F1 under controlled TTS conditions collapses to near-random on all naturalistic corpora tested. This is the expected and scientifically informative outcome for a first controlled study—it establishes that the CADD taxonomy is learnable in principle while precisely quantifying what is missing from current affective corpora. The Gross-model labels remain interpretive anchors, not psychological claims; real-speech validation is essential future work. The LOVO evaluation (mean F1 0.655) provides limited but positive evidence for voice-independent generalisation across the three available synthetic voices.

We introduced CADD to formalize the authenticity gap in digital journaling, contributed CADD-Journal as a controlled proxy corpus, and proposed DACM as a baseline architecture. Most importantly, the cross-corpus evaluation outlines a concrete research agenda: future work should construct naturalistic corpora targeting self-reflective monologues with independent dual-channel annotation and prospective user-consented labelling. By providing the task formulation, proxy dataset, baseline model, and this methodological roadmap, we aim to facilitate research into genuine cross-modal emotion sensing in the wild.

8 Limitations

We make our limitations explicit so that subsequent work can target them directly.

Construct validity of TTS-rendered dissonance. CADD-Journal isolates acoustic–lexical valence mismatch but does not instantiate genuine emotion regulation. ElevenLabs emotion tags impose a controlled mapping between intended acoustic valence and prosodic realisation; this is a necessary proxy for studying the signal in a controlled setting, but the resulting acoustic patterns may differ in distribution from suppression or reappraisal in human speakers. The Gross-model labels in Table 1 are interpretive anchors, not psychological claims.

Silver labels in cross-corpus evaluation. For CMU-MOSEI and IEMOCAP we assigned dissonance labels via SiEBERT (text sentiment) combined with an emotion-to-valence mapping (acoustic valence). The disagreement between these two estimators is itself noisy: a portion of the cross-corpus performance drop in Table 4 is therefore attributable to label noise rather than to model failure. Disentangling the two requires gold dissonance labels, which do not yet exist for these corpora.

Voice and language diversity. CADD-Journal uses three synthetic English voices. The LOVO result (mean F1 0.655) provides limited evidence for voice-independent generalisation but cannot speak to broader demographic variation (age, accent, gender presentation) or to non-English prosodic systems. The CH-SIMS result reflects both domain and language transfer and so cannot be uniquely attributed to either.

No user study of ReflectJournal. ReflectJournal is presented as a design probe and deployment platform; we have not yet evaluated whether the mismatch score S , the $S > 0.05$ surfacing threshold, or the class-specific prompts produce useful self-reflection or, conversely, induce distress in users. The threshold itself was chosen heuristically and has not been calibrated against human judgement.

Seed count and statistical power. All ablation results are reported over three seeds ({42, 123, 456}); the resulting standard deviations yield wide confidence intervals, and small differences such as DACM vs. DACM-noDIM (+0.033) should be interpreted as suggestive rather than firmly established.

9 Safe and Responsible Innovation Statement

ReflectJournal infers emotional states from speech, carrying inherent risks of misinterpretation and user distress if detections are treated as diagnostic verdicts. The system is designed exclusively for voluntary self-reflection in non-clinical populations and must not substitute for professional mental health support. CADD-Journal is constructed from synthetically generated audio; no personally identifiable data is collected or processed. Dissonance detections are framed explicitly as reflective starting points, not clinical diagnoses. Future deployments should prioritise privacy-preserving on-device inference, establish clinical guidelines distinguishing therapeutic benefit from potential harm, and evaluate cross-cultural robustness, as prosodic emotional norms vary substantially across languages and cultures.

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